METHODS for EVALUATING DIGITAL HEALTH INTERVENTIONS

Digital health interventions projects:

- explore basic questions as to whether intervention addresses identified needs
  including ← technical functionality
  feasibility

- assessment of user satisfaction

- evaluate effectiveness attributable impact

- value for money

Monitoring
Continuous process ← collecting + analising → DATA → compare results vs. expectations

in DH, process monitoring ← routine collection
review
analysis of data

measures implementation

WHO → Six stages of intervention maturity life cycle

1) preprototype
2) prototype
3) pilot
4) demonstration
5) Scale up
6) integration / sustainability

Stages to evaluating DH interventions:

Feasibility — DH system works as intended in a given context?

↓

Usability — is it used as intended?

↓

Efficacy — does it achieve intended results in a research or controlled setting?

↓

Effectiveness — Same but in non-controlled setting such as hospital, community clinic

↓

Implementation Research — assess uptake + sustainability } in given context

relies on public policy and practice

read chapter 4 WHO

Evalu
relevant re

establ
the c
ensure

RCT

Quasi-

Observat

Descrip

Evaluation Strategy

relevant research questions include
- define problem
- likely benefit

- establish the likely reach and uptake of the intervention
- the causal model describing how the intervention $\xrightarrow{\text{will reach}}$ intended benefit
- estimate overall benefit
  - effectiveness
  - costs
  - harms

RCT
- gold Standard
- only when :
  - intervention + delivery package are stable
  - implemented with high fidelity
  - reasonable likelihood of clinical effectiveness

Quasi - experimental (non randomised)

aim to → demonstrate causality

not randomised → can be biased

→ interrupted time series methods

Observational studies → can not demonstrate causation

Descriptive
- → who / when / where
- 2 main types
  - individuals
  - population

may include qualitative

RCTs - any ≠ in results (group A, group B) is attributed to the intervention

1) → VALIDITY

Assess validity [Randomisation
Allocation concealment → prevents bias until very moment of assignment

Statistical power → ability to detect a ≠

determined by
- frequency of outcome being studied
- magnitude of effect
- study design
- sample size

2) → RELIABILITY
1) how big is the treatment effect
2) how likely results are due to chance alone
} → statistical parameters
→ 95% confidence intervals
→ magnitude of effect measures
→ p values

ICC - intracluster correlation coefficient:
ρ (RHO) relatedness of clustered data by comparing variance
within clusters vs between clusters

→ Also consider clinical significance
value of results for an individual patient
Δ effect size (group A - group B)
↓
important to make a decision
(regardless of statistical significance)

3) · Precision around estimate of intervention effects

95% → how precise → wider the interval → ↓ precision

4) Using and appraising RCT

check ≠ types

ADAPTIVE DESIGNS

⊕ informative efficient ethical

⊖ more complex

designs → RCT more flexible → utilizing results accumulating in the trial → modify trial's course in accordance with pre-specified rules

NATURAL EXPERIMENTS

↑ bias → use ≠ methods testing sensitivity analyses } as additional checks before making causal inferences

+ explicitly recognize assumptions

→ TO ↑ validity

Babylon example → to read

Nice → Evidence Standards framework for DH technologies

evidence should be available or developed for < effectiveness, economic impact

not so useful for < DH private, adaptive algorithm

ECONOMIC EVALUATION

1) - 2 parameters → cost
→ outcome

essential features → cost + outcomes are analyzed
→ more than 1 strategy must be compared

5 main economic evaluation methodologies

- Cost effectiveness analysis — technical efficiency
- " utility " — quality / quantity — QALY of life
- " benefit " — economic efficiency
- " consequence " — all direct / indirect costs
- " minimisation " — only inputs

form of

measurement of outcome determines which

see again last video

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